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Hawk–Dove Evolutionary Game: Final Project Report

The Hawk–Dove game models how aggressive and peaceful behaviors in animals evolve in a population competing for a resource. The animals used in this model are Hawks (Agressive) and Dove(peaceful). This table shows the equations used to know the result of each interaction.

	Hawks	Doves
Hawks	(V-C)/2	V
Dove	0	V/2

Where V is the value of the resource, and C is the coast of injury in a Hawk vs. Hawk fight.

Equation for fitness of both types of animals:

$$\text{Hawk: } f_H(x) = x\,(V - C)/2 + (1 - x)V$$

$$\text{Dove: } f_D(x) = (1 - x)V/2$$

$$\text{Average population fitness: } f(x) = xf_H(x) + (1 - x)f_D(x)$$

The replicator equation shows how x changes with this equation.

$$\text{Xdot} = x(f_H - f)$$

$$\text{Xdot} = x(1 - x)(f_H - f_D)$$

This is the ODE I simulated in the project.

Numerical solver

I implemented both Forward Euler and RK4. Euler is used for the main tasks, and RK4 is used for the bonus 3-strategy extension.

The solver integrates the following equation over the interval t = 0, t=30 with step size h = 0.01 for accuracy.

Equation: Xdot = $x(1 - x)(f_H - f_D$

Newton's Method

To find equilibrium I solved $g(x) = x = 0$, using the Newton's method with the derivative $g'(x) = dxdot/dx$.

Stability Analysis

I used the sign of $g'(x)$ to determine whether it was stable or unstable. If the slop was negative, it was stable, and if it was positive it was unstable.

Parameter Regime Heatmap

I simulated long-run behavior from the three initial conditions. Each pair (V, C) was classified as all Dove, all Hawk, mixed ESS, or pathological.

Bonus: Three-Strategy System

Bourgeois behaves as Hawk when classified as the owner and Dove, when intruder. I implemented the replicator system and simulated the dynamics.

Results

To see how the fraction of Hawks changes over time, I simulated the replicator equation starting from five different initial values, 0.05, 0.30, 0.50, 0.80, and 0.95. All five curves converge to the same value, which matches the theoretical equilibrium of ⅓. This shows that no matter how many Hawks the population starts with, the system always settles at the same stable mix of Hawks and Doves.

This behavior is shown in Figure 1 titled Hawk–Dove Replicator. Next, I plotted the phase line xdot versus x to visualize where the equilibria occur and whether they are stable or unstable. The phase line crosses zero at, x=0, x=⅓, and x=1. The slope of the curve tells us that at x=0 and x=1 the stability is unstable because the slope is positive. The slop at x=⅓ is negative so the stability is stable. The interior equilibrium is the only stable one. This is shown in Figure 2 titled Phase Line.

I then used Newton's method to confirm the equilibrium points numerically. Starting from three different guesses, the method converged to the same three values, 0, ⅓, and 1. This matches the analytical results and shows that the numerical solver and derivative were implemented correctly.

I created a parameter regime heatmap to classify the long-run behavior for different values of V and C. The heatmap shows a large mixed ESS region when C is greater than V, an all-Hawk region when V is large relative to C, and a thin pathological strip where the system behaves irregularly. This is shown in Figure 3 titled Regimes.

For the bonus task, I added a third strategy called Bourgeois, which behaves as Hawk when it is the owner and Dove when it is the intruder. I simulated the three-strategy replicator system and found that Bourgeois eventually takes over the entire population. Hawk and Dove both decline to zero while Bourgeois rises to one. This demonstrates that Bourgeois is the ESS in the extended game. This behavior is shown in Figure 4 titled Three-Strategy Bourgeois.

Discussion and Conclusion

This project demonstrates how evolutionary game dynamics can be modeled using simple numerical tools. The Hawk–Dove game shows that aggressive behavior does not always dominate; instead, the stable outcome is a mixed population where the fraction of Hawks is determined by the ratio V/C.Newton's method found all three equilibria. Stability analysis confirmed that only the interior equilibrium is stable. The heatmap showed how changing V and C shifts the long-run behavior. Overall, the simulations and analysis support the idea that natural selection favors strategies that balance aggression and retreat, rather than pure Hawk or pure Dove behavior.